GRADUATION PROJECT · 2026

COMPUTER SCIENCE · STOCK MANAGEMENT · PREDICTIVE ANALYTICS

Intelligent Stock Management with *Predictive Product Selection for Profit Maximization*

Abdulbosit Mukhitdinov · Nurmuhammad Turgunov
Supervisor: Ibragim Atadjanov

Central Asian University
School of Engineering · Computer Science

CAU CENTRAL ASIAN UNIVERSITY

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ABSTRACT

A full-stack, role-based stock management platform for multi-location retail operations, enhanced with an XGBoost predictive engine for product selection and profit maximization. The system replaces manual spreadsheet-based tracking with real-time digital workflows across three dedicated role dashboards — Owner, Warehouse Staff, and Branch Staff — secured by PostgreSQL Row Level Security and Google OAuth. It covers the complete stock lifecycle including intake, transfers, point-of-sale, loans, returns, and auditing. The ML model, trained on 26 engineered features, predicts optimal product demand per location, enabling data-driven restocking decisions that minimize overstock costs and maximize revenue potential.

01 INTRODUCTION

Multi-branch retail businesses rely on spreadsheets and manual processes, causing stock discrepancies, data loss, and delayed decisions. Existing systems lack role-specific access and use naive forecasting — leading to simultaneous overstock and stockouts.

02 PROJECT OBJECTIVES

1. **Role-Based Access**
   Dedicated dashboards for Owner, Warehouse & Branch Staff with RLS-enforced data boundaries.
2. **Full Inventory Lifecycle**
   Intake, transfers, POS, loans, returns, corrections & auditing in one unified system.
3. **ML Demand Forecasting**
   XGBoost model outperforming the WMA baseline with per-product, per-location predictions.
4. **Real-Time & Multilingual**
   Live Supabase updates with EN, RU, UZ-Latin & UZ-Cyrillic interfaces.

03 METHODOLOGY

PHASE 1 — DESIGN & DATABASE
Field analysis of retail workflows. PostgreSQL schema with RLS, Google OAuth & atomic stored procedures to prevent race conditions.

PHASE 2 — CORE SYSTEM & DASHBOARDS
Full transaction lifecycle, inter-branch approval workflows, three role dashboards, analytics, auditing & four-language support.

PHASE 3 — ML INTEGRATION
XGBoost trained on 26 engineered features, served via FastAPI with vectorized batch inference & WMA/ML toggle.

PHASE 4 — ADVANCED FEATURES
Inventory audit system, Smart Restock engine, real-time analytics with branch comparison charts & automated Excel report generation.

04 RESULTS

KEY ACHIEVEMENT
A reactive, paper-based operation becomes a proactive, data-driven business — real-time inventory control and ML-guided restocking, available to every role, in every language.

05 PROJECT OUTPUT

WEB APPLICATION
Production-ready SPA with three role dashboards — Owner, Warehouse Staff, Branch Staff — secured by PostgreSQL Row Level Security and deployed on Vercel.

ML PREDICTION SERVER
FastAPI + XGBoost inference engine with vectorized batch predictions and automatic WMA fallback when offline.

INVENTORY MANAGEMENT SYSTEM
Seven transaction types covering the full lifecycle: intake, sale, return, loan, loan return, transfer, and correction.

06 KEY CHALLENGES

RACE CONDITIONS IN STOCK OPERATIONS
Resolved with PostgreSQL stored procedures using *SELECT ... FOR UPDATE* row-level locking, ensuring atomic deduct & add operations under concurrent users.

SUPABASE 1,000-ROW API LIMIT
Built a custom *fetchAll()* utility that auto-paginates via *.range()*, fetching all records in batches and concatenating results seamlessly.

ML BATCH PREDICTION TIMEOUT
Refactored from per-product loops to vectorized single-call inference — reducing prediction time from 30+ seconds to ~3 seconds for all products.

07 CONCLUSION

This project delivers a full-stack, role-based WMS that replaces manual workflows at a real multi-branch retail business in Uzbekistan with a real-time digital platform.

Role-enforced access control secures financial data while the ML forecasting layer transforms restocking from reactive to proactive — reducing stockouts and overstock simultaneously.

CORE CONTRIBUTION
A unified platform covering the complete warehouse lifecycle, with role-based data security and intelligent demand forecasting — purpose-built for real multi-branch retail operations.

08 RECOMMENDATION & FUTURE WORK

Mobile Application
Native app with barcode & QR scanning for floor-level stock counts and sales operations.

Supplier Auto-Reordering
Auto-generate purchase orders when ML predicts stock below threshold — closing the prediction-to-procurement loop.

Advanced Analytics
Dead stock identification, ABC classification, seasonal heatmaps & scheduled email report delivery.

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